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Spider-Scents: Grey-box Database-aware Web Scanning for Stored XSS

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As web applications play an ever more important role in society, so does ensuring their security. A large threat to web application security is XSS vulnerabilities, and in particular, stored XSS. Due to the complexity of web applications and the difficulty of properly injecting XSS payloads into a web application, many of these vulnerabilities still evade current state-of-the-art scanners. We approach this problem from a new direction—by injecting XSS payloads directly into the database we can completely bypass the difficulty of injecting XSS payloads into a web application. We thus propose Spider-Scents, a novel method for grey-box database-aware scanning for stored XSS, that maps database values to the web application and automatically finds unprotected outputs. Spider-Scents reveals code smells that expose stored XSS vulnerabilities. We evaluate our approach on a set of 12 web applications and compare with three state-of-the-art black-box scanners. We demonstrate improvement of database coverage, ranging from 79% to 100% database coverage across the applications compared to the range of 2% to 60% for the other scanners. We systematize the relationship between unprotected outputs, vulnerabilities, and exploits in the context of stored XSS. We manually analyze unprotected outputs reported by Spider-Scents to determine their vulnerability and exploitability. In total, this method finds 85 stored XSS vulnerabilities, outperforming the union of state-of-the-art's 32.

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